

Technical Section

Choreographing multi-degree of freedom behaviors in large-scale crowd simulations

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ABSTRACT

This study introduces a novel framework for choreographing multi-degree of freedom (MDoF) behaviors in large-scale crowd simulations. The framework integrates multi-objective optimization with spatio-temporal ordering to effectively generate and control diverse MDoF crowd behavior states. We propose a set of evaluation criteria for assessing the aesthetic quality of crowd states and employ multi-objective optimization to produce crowd states that meet these criteria. Additionally, we introduce time offset functions and interpolation progress functions to perform complex and diversified behavior state interpolations. Furthermore, we designed a user-centric interaction module that allows for intuitive and flexible adjustments of crowd behavior states through sketching, spline curves, and other interactive means. Qualitative tests and quantitative experiments on the evaluation criteria demonstrate the effectiveness of this method in generating and controlling MDoF behaviors in crowds. Finally, case studies, including real-world applications in the Opening Ceremony of the 2022 Beijing Winter Olympics, validate the practicality and adaptability of this approach.

1. Introduction

Crowd behavior planning has been widely applied in various fields, such as large-scale crowd performances [1], unmanned airborne vehicles (UAVs) choreography [2], crowd evacuation [3], and in the production of films and video games. Large-scale crowd performances are particularly important in major events such as Olympic opening ceremonies and national celebrations. However, designing and executing a large-scale crowd performance with precision and coordination presents significant challenges for both directors and performers. The necessity for spatial regularity and synchronization of individual actions at the macro level, combined with the high dimensionality of individual behaviors at the micro level, makes crowd performance planning extremely complex. Consequently, their spatial and temporal coordination becomes very difficult. Traditional approaches depend on the director's intuition and experience, iterative design refinements, and methods involving real actor choreography, which are often time-consuming and costly.

Recently, crowd simulation methods have been utilized to address the behavior planning challenges in large-scale crowd performances, improving the efficiency of choreography and rehearsal processes, reducing human costs, and expanding the scope of crowd performance

planning possibilities. However, current crowd simulation methods primarily focus on individual steering and speed decisions, often overlooking multi-degree of freedom (MDoF) behaviors beyond these aspects, such as changes in height during squatting, body joint movements, and the manipulation of objects by individuals, which can form complex behaviors of regular performance movements. Current action generation methods predominantly concentrate on skeletal movement studies, ignoring interactions and collaborations among individuals within the crowd, as well as the diversity of crowd behaviors that emerge from such interactions.

To expand the dimensions of crowd behaviors and enhance the diversity in behavior planning, we propose an interactive framework that combines multi-objective optimization with spatio-temporal interpolation. This framework is specifically designed to generate and control MDoF behaviors in performing crowds. Typically, planning crowd performance behaviors involves two key stages. In the first stage, the design focuses on creating static formations that adhere to aesthetic principles such as coordination, continuity, and complexity. During this stage, users provide interactive inputs as predefined constraints, and intelligent optimization methods are employed to generate static crowd states that meet the specified evaluation criteria. The second

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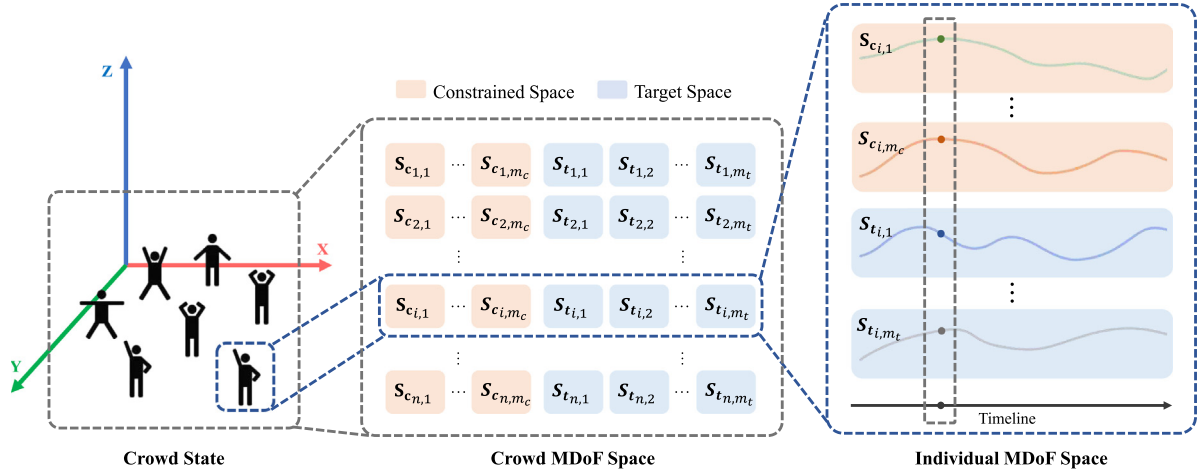


Fig. 1. Modeling of MDoF behavior in crowds. On the left is the behavior state of the crowd in real three-dimensional space, in the middle is the matrix representation of the crowd's MDoF state space, and on the right is the temporal variation of individual MDoF behavior space. In the middle and right parts, the orange solid boxes represent the constrained space, which is the behavior state pre-defined by experts or data-driven, and the blue solid boxes represent the target behavior space, which is the behavior space that our method needs to optimize and control.

stage focuses on generating dynamic transitions between different static states. In performance behaviors, state transitions for each individual in the crowd often involve time delays. Our framework employs spatio-temporal interpolation and configurable control functions to create diverse dynamic interpolated states for the crowd.

Additionally, effective human-computer interaction methods are crucial for planning crowd behaviors. To this end, we have developed tools such as a sketching brush and custom spline curves, which enable users to interactively adjust crowd behaviors. We also validated the expandability of our method through quantitative experiments involving various configurations of aesthetic criteria, applied to different performance formations and behaviors. Furthermore, we successfully applied the proposed method to the planning and rehearsal of real-world crowd performances, thereby demonstrating its practicality and scalability.

The main contributions of this paper include:

1. Modeling the MDoF behavioral space of crowds, introducing the concepts of constrained and target behavioral spaces, and proposing a framework for planning MDoF crowd behaviors based on multi-objective optimization and spatio-temporal ordering.
2. Proposing three aesthetically aligned evaluation criteria for MDoF crowd behavior states: local continuity, mutual information of state distribution, and entropy of behavior states.
3. Developing time offset functions and interpolation progress functions to adjust and control dynamic transitions of crowd states, along with interactive design tools like sketching and spline curves.

The subsequent sections of this paper are organized as follows: we first introduce the Related Work in 2, followed by Section 3 where we present the modeling of the MDoF behavioral space of crowds and the implementation of the modules in our proposed framework. Section 4 discusses the quantitative experimental results of optimizing crowd states using this method, the visualization of the crowd state generation and control processes, and case studies of the method's application. Finally, Section 5 will conclude our study and propose future work prospects.

2. Related work

2.1. Crowd simulation

Existing crowd simulation methods primarily focus on simulating the turning and speed decisions of a crowd. For instance, macroscopic

models [4], used in large-scale, dense scenarios, treat the crowd as a unified, continuous entity with movement controlled by potential fields or fluid dynamics [5,6]. In microscopic models [4], individuals are considered discrete objects, with their movements influenced by neighbors and obstacles, as seen in the Social Force Model (SFM) [7] and Optimal Reciprocal Collision Avoidance (ORCA) algorithm [8]. These methods can achieve collision avoidance and can be combined with path planning [9] to form realistic simulations of crowds.

To ensure aesthetic qualities in crowd behavior planning, Alonso-Mora et al. [10] and Zheng et al. [11] have respectively applied Centroidal Voronoi Tessellation (CVT) to crowd formation changes, thus creating visually appealing keyframes. Xu et al. [12] used dynamic crowd mutual information to maintain the smoothness and stability of crowd movements. Zheng et al. [2] employed the 2-Wasserstein distance to generate geometrically coherent intermediate states in UAVs formation transitions.

Moreover, appropriate human-computer interaction methods are crucial for authoring diverse crowd behaviors [13]. For example, Ulicny et al. [14] designed Crowdbrush, a tool spraying in virtual 3D scenes that allows users to distribute, modify, and control crowd agents with instantaneous visual feedback. He et al. [15] proposed a method for formation control in crowds through interactive gesture recognition that matches crowd formation templates. Kim et al. [16] introduced a cage-based method for controlling and editing crowd animations. Colas et al. [17] defined Interaction Fields (IFs), using interactive sketches to control local behavior templates in crowds. Furthermore, methods combining interactive sketches with navigation meshes [18] or vector fields [6] have been developed to control crowd behaviors.

In our work, we have extended beyond traditional steering and speed decision-making to plan MDoF crowd behaviors. Furthermore, inspired by the previous work on CrowdBrush [14] and Spline techniques [19], we also developed intuitive interactive tools for planning MDoF crowd behaviors.

2.2. Motion generation

The current research on individual movement has achieved significant progress, where researchers have generated or reconstructed individual motions through various interactive interfaces and different data inputs. For example, Zhang et al. [20] used diffusion models to generate corresponding action sequences from textual descriptions. Zhuang et al. [21] employed autoregressive generative models to create individual dance movements based on musical styles, rhythms,

and melodies. Chen et al. [22] proposed a music-driven dance motion synthesis system for choreography, while Ponton et al. [23] developed the SparsePoser framework, capable of reconstructing accurate individual actions from sparse trackers.

Quantitative computational methods can be employed for motion encoding or evaluating aesthetic-related features [24]. For example, Anjos et al. [25] introduced a novel three-dimensional visualization of dance movement qualities, using colored point-cloud videos resembling heatmaps to visualize movement qualities such as movement direction and synchronization between dancers. Piana et al. [26] created a multimodal repository for dance movement qualities, focusing on three custom expressive qualities: Smoothness, Impulsiveness, and Rigidity.

Building on the foundation of individual movement research, our work expands to explore the behavioral collaboration among individuals within a crowd. Additionally, We also utilized mutual information and spatial continuity as evaluation criteria for aesthetic features.

2.3. Optimization algorithm

Intelligent optimization algorithms have made significant progress in solving complex and high-dimensional problems [27]. For example, Duan and Qiao [28] used genetic algorithms for robot path planning, Zhang et al. [29] employed swarm intelligence algorithm for UAVs path planning. Additionally, optimization algorithms are employed for artistic purposes, including generating interactive designs [30] and choreography plans [31–33]. Beyond their ability to find optimal solutions, the randomness inherent in these algorithms is also a helpful feature for generating unseen planning schemes [34]. Generating crowd behavioral states often involves multiple heterogeneous evaluation criteria, making multi-objective methods necessary [28,35]. Pareto optimality [36] is one of the most popular approaches for addressing these multi-objective problems. To generate diverse crowd behavioral states, we have selected the Multi-Objective Grey Wolf Optimizer [37] as our method, which allows for the exploration of various potential solutions, balancing between different objectives to create a wide range of crowd behaviors.

3. Methodology

In this section, we will first model the MDoF behaviors of a crowd and then propose a framework for planning these behaviors based on optimization and spatio-temporal ordering. We have developed several aesthetic-related evaluation criteria for crowd behavior states, combined with user-input state values as target functions, and optimized them using metaheuristic optimization algorithms. We have designed various control functions for interpolating crowd states, thereby enabling dynamic control of the crowd's behavior. Additionally, we have developed a user input module that allows users to intuitively and flexibly adjust MDoF behaviors through several interactive methods. All the key mathematical symbols used in this section are listed in Table 1.

3.1. MDoF behavior space modeling

The behavior of crowds varies across different contexts. For instance, in the context of pedestrian crowds, their behavior primarily involves speed adaptation to perform local steering in order to move towards a goal while avoiding obstacles. In contrast, in performance crowds, the degrees of freedom of behaviors are broader and include not only spatial adjustments of position and direction but also the angles of limb joints, the height of squats, and the status of any handheld props.

We model the collective behavior of crowds during performances and other regular activities as a MDoF behavior state space, represented by the matrix $S \in \mathbb{R}^{n \times m}$, where n is the number of individuals in

the crowd, and m is the number of degrees of freedom in the behavioral space, as shown in Fig. 1. During regular behavior, individuals' motions are often constrained in some dimensions while exhibiting regular motion in others. The constrained behavioral space is denoted as $S_c \in \mathbb{R}^{n \times m_c}$, where m_c is the number of constrained dimensions. These constraints can be manually defined by experts or controlled through data-driven methods. The freely moving behavioral space, which we defined as, the target space of degrees of freedom, is represented by $S_t \in \mathbb{R}^{n \times m_t}$, where m_t is the number of dimensions available for planning. The target space of degrees of freedom is optimized and controlled by our proposed framework. The MDoF behavioral space of a crowd is a continuous state space, with each individual's constrained behavioral states $\{S_c(i, k) \mid 1 \leq i \leq n, k \leq m_c\}$ and target behavioral states $\{S_t(i, k) \mid 1 \leq i \leq n, k \leq m_t\}$ changing continuously over time. The static state at a particular moment is represented as $S(t)$, where t denotes the timestamp. The relationship within the behavior state space is as follows:

$$S = [S_c \oplus S_t] \quad (1)$$

3.2. Overall framework

To facilitate the generation and control of MDoF behaviors in crowds, we have developed an interactive planning framework for crowd behaviors based on optimization and spatio-temporal ordering. This framework comprises three main modules:

1. Crowd static MDoF states optimization module (Module A in Fig. 2): This module includes a multi-objective intelligent optimization algorithm and a set of target optimization functions based on behavioral evaluation criteria. It takes in user-inputted target behavior evaluation feature values and, upon optimization convergence, generates static crowd states that meet these evaluation criteria.
2. Crowd MDoF states interpolation module (Module B in Fig. 2): This module receives the static crowd states generated by Module A, along with user-defined spatial ordering values, time offset functions, and interpolation progress functions. It then uses spatio-temporal ordering algorithms to produce diverse transitional and cyclic dynamic crowd behavior states with temporal variations.
3. User interact module (Module C in Fig. 2): In this module, we have designed a user interaction graphical interface (GUI), enabling users to adjust MDoF crowd behavior states through numerical inputs, sketching brush, and custom spline curve.

The implementation details of each module will be described in the following sections.

3.3. Crowd static MDoF states optimization module

This module is focused on generating static states for crowds. It combines the states of the constrained behavioral space of the crowd and uses the difference between predefined aesthetic evaluation features and user-inputted target feature values as the objective function. A multi-objective intelligent optimization algorithm is employed to iteratively optimize the randomly initialized target states of the crowd. This results in a set of non-dominated Pareto optimal solutions, yielding diverse static crowd states that satisfy both constraint adjustments and target functions.

In terms of framework design, we adhere to the principle of the No Free Lunch (NFL) theorem [38], which suggests that no single optimization algorithm is best for all tasks. Consequently, any algorithm capable of multi-objective optimization can be applied within this module. For our practical implementation, we have utilized the Multi-Objective Grey Wolf Optimizer (MOGWO) [37]. The subsequent sections will first define the optimization problem for static MDoF states in crowds.

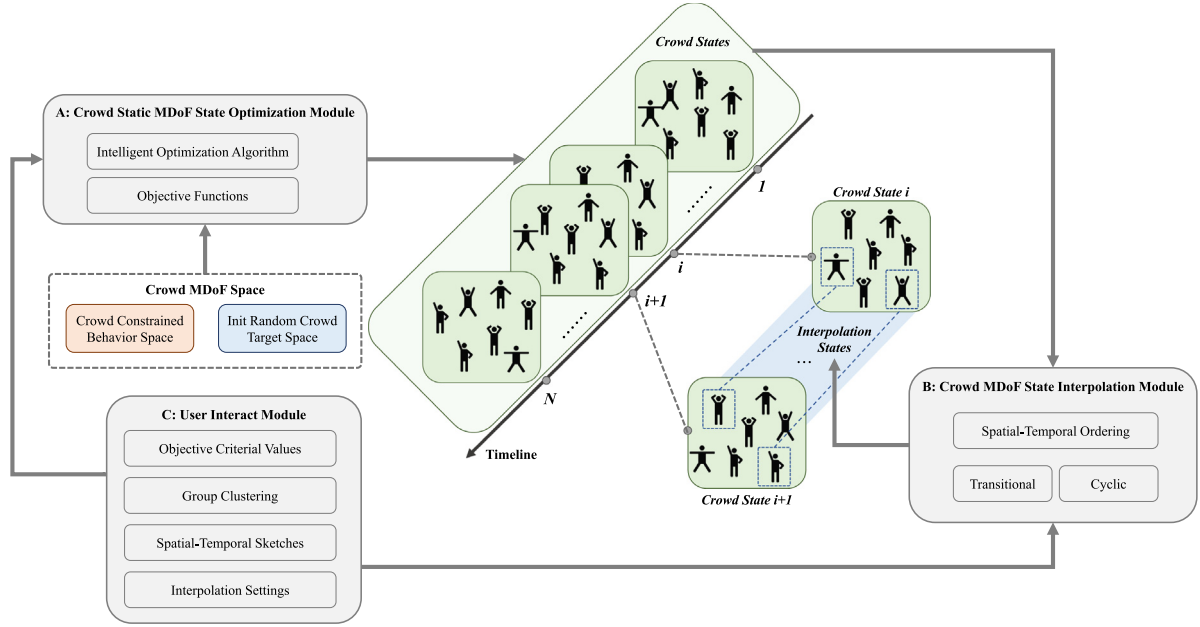


Fig. 2. MDof behavior planning framework based on multi-objective optimization and spatio-temporal ranking. In this figure, Module A represents the crowd static MDof state optimization module, capable of generating stationary states that align with target evaluation criteria. Module B represents the crowd MDof state interpolation module, which produces transitional states between adjacent static crowd states using grouping clustering and spatio-temporal ranking. Additionally, Module C is the user input module, allowing users to provide behavior criteria, either manually defined or data-driven, for the first two modules.

Table 1
Key mathematical symbols in crowd MDof behavior modeling.

Symbol	Description
S	Crowd behavioral space
S_c	Constrained behavioral space of a crowd
S_t	Target behavioral space of a crowd
s	MDof vector of an individual
s_i	MDof vector of the i individual in a crowd
$s(j)$	The state vector of j degree of freedom in a crowd
$s_{i,j}$	The state vector of j degree of freedom of the i individual in a crowd
$s'_{i,j}$	The normalized state vector of j degree of freedom of the i individual in a crowd
$p_k(s(j))$	The margin probability of the j degree of freedom in the k bin

Then, we will outline the crowd state evaluation criteria and target functions based on aesthetic characteristics. Finally, the application of the Multi-Objective Grey Wolf Optimizer [37] in optimizing crowd behavior states will be discussed.

3.3.1. Problem definition

For the optimization of static MDof states in crowd behaviors, considering the characteristics of collective crowd behaviors and aesthetic evaluation standards in choreography, the following hypotheses can be proposed:

1. Certain behavioral spaces of the crowd are subject to constraints. For instance, the position of each individual within the crowd might be fixed, their trajectories predetermined, or their orientations set in advance.
2. The behavior state (their animation, orientation, etc.) of each individual in the crowd is associated with their spatial position.
3. The behavior state of each individual in the crowd is related to the states of other individuals within their vicinity.
4. The state formed by the crowd adheres to certain aesthetic evaluation standards, such as harmony, continuity, and complexity.

Multi-objective optimization refers to the optimization of problems with multiple objective functions under multiple constraints. The optimization of the static state of a crowd's MDof behavior can be

formulated as the following minimization problem:

Minimize:

$$J(S) = J_1(S), J_2(S), \dots, J_o(S)$$

Subject to:

$$g_i(S) \geq 0, i = 1, 2, \dots, n_g$$

$$h_i(S) = 0, i = 1, 2, \dots, n_h$$

$$L_j \leq S_i(i, j) \leq U_j, i = 1, 2, \dots, n, j = 1, 2, \dots, m_t$$

(2)

Within this framework, o denotes the number of objective functions, n_g is the number of inequality constraints, n_h is the number of equality constraints, and m_t represents the number of target degrees of freedom. The value boundaries for the i target degree of freedom are given by $[L_i, U_i]$. This forms the general optimization framework for a crowd's MDof state, where the input for the objective function $J_i(S)$ includes all dimensions of the crowd's behavioral state, both constrained and target behavioral spaces. This is because the evaluation criteria for the target behavioral space may be related to the constrained behavior space. Both equality and inequality constraints can be driven by external data or user specifications, and both objective functions and constraints can be tailored to different crowd behaviors. Additionally, each target degree of freedom within the behavioral space has value boundaries, such as the spatial range of an individual, the range of orientation angles, etc.

3.3.2. Objective functions

This section introduces several objective functions for optimization based on the aesthetic evaluation criteria of a crowd's MDoF static state.

Mutual information. Mutual Information (MI) is a well-known concept in the field of information theory, designed to quantify the mutual dependence between two random variables. Generally, the MI of two discrete sequence variables, G and H, can be defined as follows:

$$I(G, H) = \sum_{i,j} p(g_i, h_j) \log_n \left(\frac{p(g_i, h_j)}{p(g_i)p(h_j)} \right) \quad (3)$$

where $p(g_i)$ and $p(h_j)$ represent the individual distributions of G and H, respectively, while $p(g_i, h_j)$ represents the joint probability distribution of G and H.

By adjusting the target mutual information between the numerical distributions of different dimensions in the crowd's behavioral space, we can control the correlation between crowd behaviors in different dimensions. For instance, by modifying the mutual information value between the target behavioral space and the constrained behavior space, the state of the crowd in the target behavioral space can be controlled based on the distribution in the constrained space. When calculating the state distribution between different dimensions of crowd behavior, the data in the behavioral space is typically normalized first. The normalization formula is as follows:

$$s'_{i,j} = \frac{s_{i,j} - \min(s(j))}{\max(s(j)) - \min(s(j))} \quad (4)$$

where $\max(s(j))$ and $\min(s(j))$ represent the normalization boundaries of the crowd's values in the j degree of freedom.

Subsequently, we calculate the mutual information using the normalized values of the degrees of freedom. Since the state of the crowd is composed of multiple individuals, the distribution of each degree of freedom behavior is discrete. Therefore, we divide the value interval of each normalized degree of freedom behavior into K equal bins, with the default value of $K = 20$. The calculation of the marginal probability of the j degree of freedom in the k bin is as follows:

$$p_k(s(j)) = \frac{1}{n} \sum_{1 \leq i \leq n} s'_{i,j} \in \left[\frac{k-1}{K}, \frac{k}{K} \right] \quad (5)$$

Based on Eq. (5), the mutual information $I(s(j_1), s(j_2))$ between the states of the j_1 and j_2 degrees of freedom behaviors in the crowd can be calculated as follows:

$$I(s(j_1), s(j_2)) = \sum_{k_1, k_2} p_{k_1, k_2}(s(j_1), s(j_2)) \log_2 \left[\frac{p_{k_1, k_2}(s(j_1), s(j_2))}{p_{k_1}(s(j_1)) p_{k_2}(s(j_2))} \right] \quad (6)$$

To allow for the adjustment of the crowd's MDoF behavioral states by user input, the minimization objective function set by mutual information between the j_1 and j_2 degrees of freedom behavioral states is as follows:

$$J_{MI_{j_1, j_2}}(S) = \left| C_{mi_goal}(j_1, j_2) - I(s(j_1), s(j_2)) \right| \quad (7)$$

where $C_{mi_goal}(j_1, j_2)$ represents the target mutual information value between the j_1 and j_2 degrees of freedom behavioral states, received from the user interaction module. Throughout the entire optimization process, based on the specific MDoF behavioral states of the crowd to be generated, the mutual information objective function for different degrees of freedom can be set as one or multiple targets.

Spatial continuity. In a crowd behavior space with aesthetic qualities, the behavioral difference between an individual and its neighboring individuals (neighbors) typically does not vary significantly, exhibiting a certain extent of smoothness. Considering this characteristic, we propose an evaluation metric for spatial continuity.

We define the set of all neighbors of individual i within a radius R as N_i , with the definition as follows:

$$N_i = \{j \mid D(s_i, s_j) < R, j \neq i, G(i) = G(j)\} \quad (8)$$

To achieve a more diverse range of crowd behaviors, each individual is assigned to a group. $G(i)$ represents the group label of the i individual. An individual is only added to the neighbor set if they are within the neighborhood range and share the same group label. The grouping is defined by the user interaction module and will be specifically introduced in Section 3.5.

$D(s_i, s_j)$ is a function measuring the distance between individual i and individual j . This is assessed using the Euclidean distance across dimensions that represent position information within the constrained behavioral space.

The definition of the continuity evaluation value $F_{smooth}(i, j)$ for individual i with its neighbors on the degree of freedom j is as follows:

$$F_{smooth}(i, j) = \frac{1}{|N_i|} \sum_{k \in N_i} |s_{i,j} - s_{k,j}|^2 \quad (9)$$

The definition of the spatial continuity $F_{smooth}(j)$ for the entire crowd on the degree of freedom j is as follows:

$$F_{smooth}(j) = \frac{1}{n} \sum_{i=1}^n F_{smooth}(i, j) \quad (10)$$

Combining the target continuity value for the behavioral space input by the user, the minimization objective function for assessing the smoothness of the behavioral space of the j degree of freedom is as follows:

$$J_{smooth_j}(S) = \left| C_{smooth_goal}(j) - F_{smooth}(j) \right| \quad (11)$$

where $C_{smooth_goal}(j)$ represents the target spatial smoothness for the behavior of the j degree of freedom, as input by the user interaction module. For behaviors of different degrees of freedom in the target behavioral space, multiple distinct spatial smoothness target values and corresponding objective functions can be set.

Entropy. To evaluate the uniformity or complexity of the crowd's MDoF behaviors, we utilize the concept of entropy. Shannon entropy, a measure of information, can be used to assess the diversity of solutions in optimization methods [39]. Lee et al. [31] applied entropy to measure the complexity of crowd behaviors in floor patterns within crowd choreography. In our work, entropy is employed to assess the complexity of the behavior for a particular target degree of freedom.

For the evaluation of the behavior of the j degree of freedom in the crowd, we first divide the range of its samples into K equal-sized bins, with a default value of $K = 20$. Each bin size is denoted as $\Delta s(j)$. Subsequently, we calculate the probability of each bin k ($1 \leq k \leq K$), as follows:

$$\Delta s(j) = \frac{\max(s(j)) - \min(s(j))}{K} \quad (12)$$

$$p_k(s(j)) = \frac{1}{n} \sum_{1 \leq i \leq n} s_{i,j} \in \left[\frac{k-1}{K} \cdot \Delta s(j), \frac{k}{K} \cdot \Delta s(j) \right], \quad (13)$$

Then, the entropy $H(s(j))$ of the behavioral space for the j degree of freedom can be calculated through the probability distribution:

$$H(s(j)) = - \sum_{k=1}^K p_k(s(j)) \log [p_k(s(j))] \quad (14)$$

Combining the target entropy value for the behavioral space input by the user, the minimization objective function for assessing the complexity of the behavioral space of the j degree of freedom is as follows:

$$J_{entropy_j}(S) = \left| C_{entropy_goal}(j) - H(s(j)) \right| \quad (15)$$

where $C_{entropy_goal}(j)$ is the target spatial complexity for the behavior of the j degree of freedom, as input by the user interaction module. As with the two objective functions mentioned above, for behaviors of different degrees of freedom in the target behavioral space, multiple distinct spatial complexity values and corresponding objective functions can be set.

3.3.3. Multi-objective grey wolves optimization algorithm

The Grey Wolf Optimizer (GWO) [40] is a metaheuristic optimization method primarily designed for single-objective optimization, proposed by Mirjalili et al. GWO is inspired by the hunting behavior and social hierarchy of grey wolves in nature. Similar to other metaheuristic approaches, it initializes the optimization process by generating a set of random candidate solutions. In each iteration, the top three wolves are considered the alpha (α), beta (β), and delta (δ), which guide the rest of the pack towards promising areas of the search space. The remaining grey wolves are considered omegas (ω), and these ω wolves follow the leading three in the search for the global optimum. In addition to social leadership, to simulate the encircling behavior of grey wolves during the hunt, the following equations are proposed:

$$\vec{D} = |\vec{C} \cdot \vec{X}_p(t) - \vec{X}(t)| \quad (16)$$

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot \vec{D} \quad (17)$$

where t represents time, $\vec{X}_p(t)$ denotes the position of the targeted prey, and $\vec{X}(t)$ represents the current position of the grey wolf. The coefficients $\vec{A}$ and $\vec{C}$ are utilized to adjust the random exploration behavior of the grey wolf.

$\vec{A}$ and $\vec{C}$ are defined as follows:

$$\vec{A} = 2\vec{a} \cdot \vec{r}_1 - \vec{a} \quad (18)$$

$$\vec{C} = 2 \cdot \vec{r}_2 \quad (19)$$

where the elements in vector $\vec{a}$ gradually decrease linearly from 2 to 0, while $\vec{r}_1$ and $\vec{r}_2$ are random vectors with elements in the range of [0, 1]. Consequently, the magnitude of $\vec{A}$ is a random number within the range of $[-\|\vec{a}\|, \|\vec{a}\|]$. Initially, if $\|\vec{A}\| > 1$, according to Eq. (17), the wolf moves away from the target, representing the exploration phase of the metaheuristic algorithm, which is beneficial for escaping local optima. As the number of iterations increases and $\|\vec{A}\| < 1$, the wolf approaches the target, signifying the exploitation process of the metaheuristic algorithm to accelerate convergence. The vector $\vec{C}$ remains random, playing a crucial role in avoiding local optima. The $\vec{C}$ vector can also be interpreted as the influence of obstacles in the natural environment on the approach towards the prey. Generally, obstacles in nature, appearing on the wolf's hunting path, actually prevent them from reaching their prey quickly and conveniently. This is precisely the role of the vector $\vec{C}$. Depending on the position of the wolf, it can randomly assign a weight to the prey, making it more difficult and distant for the wolf to reach the prey, and vice versa.

During the optimization process, the omega wolves (ω) update their positions surrounding the alpha (α), beta (β), and delta (δ) wolves. Therefore, each omega wolf (ω) can reposition itself relative to the alpha (α), beta (β), and delta (δ) simultaneously, as demonstrated below:

$$\begin{aligned} \vec{D}_\alpha &= |\vec{C}_1 \cdot \vec{X}_\alpha - \vec{X}|, \\ \vec{D}_\beta &= |\vec{C}_2 \cdot \vec{X}_\beta - \vec{X}|, \\ \vec{D}_\delta &= |\vec{C}_3 \cdot \vec{X}_\delta - \vec{X}| \end{aligned} \quad (20)$$

$$\begin{aligned} \vec{X}_1 &= \vec{X}_\alpha - \vec{A}_1 \cdot (\vec{D}_\alpha), \\ \vec{X}_2 &= \vec{X}_\beta - \vec{A}_2 \cdot (\vec{D}_\beta), \\ \vec{X}_3 &= \vec{X}_\delta - \vec{A}_3 \cdot (\vec{D}_\delta) \end{aligned} \quad (21)$$

$$\vec{X}(t+1) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3} \quad (22)$$

To address multi-objective optimization problems, the Grey Wolf Optimizer has been extended to a Multi-Objective Grey Wolf Optimizer (MOGWO) [37]. Similar to MOPSO [41], by integrating an archive of optimal solutions and an elite selection strategy, alpha, beta, and delta solutions can be selected from the archive to act as leaders in the hunting process.

For the optimization problem concerning the static state of the crowd's MDoF behavior, the target optimization functions are those listed in Section 3.3.2. These functions can be further expanded for different types of crowd behaviors. The solution space is the target behavioral space of the crowd, as mentioned in Section 3.1, denoted as S_t .

In multi-objective optimization problems, Pareto dominance is a key concept that defines the superiority of one solution over others. A solution is considered to Pareto dominate another if it is not worse than other solutions in all objectives and is better in at least one objective compared to the other. Since the designed objective functions $J(S_i)$ aim to minimize the difference between evaluation criteria and user input standards, we define a solution S_{i1} as Pareto dominating another solution S_{i2} , denoted as $S_{i1} < S_{i2}$, if and only if S_{i1} and S_{i2} satisfy the following conditions:

$$\begin{cases} \forall i \in \{1, 2, \dots, o\}, & J_i(S_{i1}) \leq J_i(S_{i2}) \\ \exists j \in \{1, 2, \dots, o\}, & J_j(S_{i1}) < J_j(S_{i2}) \end{cases} \quad (23)$$

The Pareto optimal solution front, also known as Pareto front or Pareto boundary, refers to the set of all solutions in the objective function space that are not Pareto dominated by any other solutions. Within this set, no solution can be improved on one objective without sacrificing at least one other objective. In relation to the optimization problem of the crowd's MDoF static state, our ultimate goal is to obtain the optimized non-dominated Pareto optimal solution front P^* , which is defined as follows:

$$P^* = \{S^* \in S_t \mid \nexists S' \in S_t : S' < S^*\} \quad (24)$$

Thus, the main essence of MOGWO can be described as determining the dominance relationships based on the objective functions for the crowd's MDoF states. The Pareto optimal solution set is then preserved in an archive. During the optimization process, the multi-objective optimal solutions are updated and the archive is maintained. The archive at the end of the optimization process contains the MDoF static states of the crowd's target behavioral space that meet the requirements.

3.4. Crowd MDoF states interpolation module

This section explains how to control the interpolation states between the static states of crowds. First, we introduce the method for interpolating the MDoF states of each individual in the crowd, focusing on spatio-temporal ordering and control through time offset functions. Next, we discuss the use of different interpolation progress functions to create two types of crowd state interpolations: transitional interpolation and cyclic interpolation. To illustrate this module, we use a prop-inclusive crowd performance as an example for planning MDoF crowd behavior, providing a visual demonstration of the process.

3.4.1. Spatio-temporal ordering

Previous research on crowd state transitions, such as formation transformation [2,15], typically involved the entire crowd undergoing state transitions simultaneously, with few methods focusing on controlling and designing the sequence of movements within the crowd. To address this gap, we propose a method based on spatial positioning within the crowd to order the timing of state transitions, referred to as spatio-temporal ordering. Each individual i in the crowd is assigned

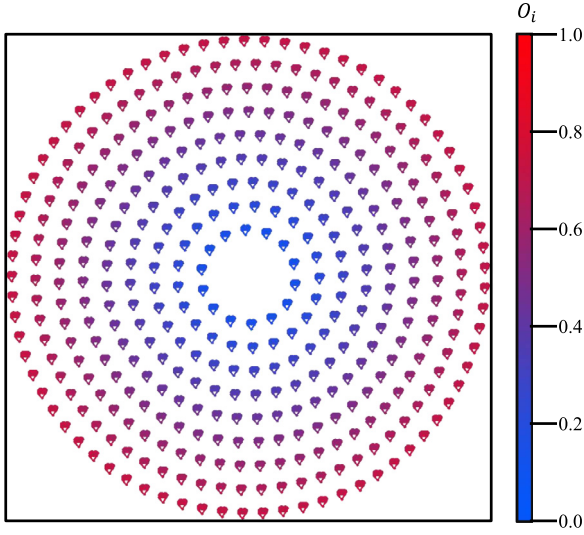


Fig. 3. The ordered result of 393 agents using $\text{comp}(S)$ comparing distance to crowd's center.

an ordering value $O_i \in [0, 1]$. O_i is generally assigned based on the individual's spatial position and can be calculated using a predefined spatial ordering function, which is defined as follows:

$$O_i = \text{Sort}(S, i, \text{Comp}(S)) \quad (25)$$

where $\text{Comp}(S)$ is a spatial ordering comparison function that determines the ordering priority of each individual based on the crowd's positional information. The smaller the value of O_i , the higher its ordering priority. For instance, Fig. 3 shows an example of $\text{Comp}(S)$ spatially ordering all individuals based on the distance function of each individual to the center point of the crowd. In addition to this, the ordering value O_i can also be obtained from more clustering distance functions in the user interaction module or from interactive sketches.

In addition, we have developed a function named $\text{TO}(O_i)$, which serves as an interpolation time offset mechanism. This function, taking the spatial ordering value as its input, is crucial for determining the commencement timing for each individual i during the process of interpolating crowd states. The formulation of $\text{TO}(O_i)$ can range from a straightforward linear function to more complex forms, customized via the curve tools provided in the user input interface. The primary role of the $\text{TO}(O_i)$ function lies in its ability to convert spatially ordered data into temporal sequences pertinent to the interpolation of crowd states, thus embodying the fundamental spatio-temporal ordering concept of this system. Different $\text{TO}(O_i)$ functions can add diversity to crowd behavior during the interpolation process and offer great extensibility.

Based on the above definitions, given the initial and target MDoF behavior states of a crowd, denoted as S^A and S^B respectively, at time t , the definition of the interpolated state $S_{i,j}(t)$ for the j degree of freedom behavior of individual i in the crowd is as follows:

$$S_{i,j}(t) = S_{i,j}^A + \text{Lerp}[\alpha(t, i)] \cdot (S_{i,j}^B - S_{i,j}^A) \quad (26)$$

$$\alpha(t, i) = \text{clamp} \left[\frac{t - \text{TO}(O_i)}{T}, 0, 1 \right] \quad (27)$$

where $\alpha(t)$ represents the interpolation progress of individual i at time t , and T is the time required for each individual to complete the state interpolation. That is, when the interpolation time t is less than the individual i 's interpolation time offset value $\text{TO}(O_i)$, the individual maintains the initial state. When the interpolation time t exceeds the time offset value, individual i starts the interpolation to switch states, thereby enabling temporal control over the transition of different individuals' MDoF states. $\text{Lerp}(\alpha)$ is a function controlling the interpolation

progress, which can expand the diversity of crowd behavior during interpolation. We will introduce it in detail in the next subsection.

3.4.2. Transitional and cyclic interpolation

$\text{Lerp}(\alpha)$ is a function that controls the progress of interpolation. It manages the dynamic distance between the initial and target states of each individual during interpolation, thereby enhancing the diversity of the crowd's behavior throughout the interpolation process. For example, it can control the magnitude and periodicity of the MDoF state transitions of the crowd during interpolation. By setting the start and end values of the $\text{Lerp}(\alpha)$ function within its domain, interpolation can be categorized into transitional interpolation and cyclic interpolation. Transitional interpolation refers to a method where the crowd smoothly transitions from one MDoF state to another, while cyclic interpolation refers to interpolation under the condition that the initial and target MDoF states are the same, thus forming a closed loop. Examples of these two types of interpolation are illustrated in Fig. 4:

The $\text{Lerp}(\alpha)$ function controls the extent of interpolation within the dashed box in Fig. 4. In order to achieve transitional interpolation, it must satisfy $\text{Lerp}(0) = 0 \wedge \text{Lerp}(1) = 1$. To achieve cyclic interpolation, the condition $\text{Lerp}(0) = \text{Lerp}(1)$ is required. The other values of the function within the interval $(0, 1)$ can be controlled either by predefined explicit functions or by spline tools in the user interaction module. In addition, for cyclic interpolation, if the number of cycles is greater than 1, the interpolation progress function $\alpha(t, i)$ can be a periodic function with respect to t , allowing multiple repeated interpolations:

$$\alpha(t, i) = \text{frac} \left[\frac{t - \text{TO}(O_i)}{T} \right] \quad (28)$$

3.5. User interact module

This section introduces the user interaction module of the framework. This module allows users to input various types of data to define MDoF behavior characteristics for the first two modules. These inputs include numerical data, group clustering, and user-defined spline functions.

3.5.1. Numerical input

The numerical input method is used to set the target evaluation values within the crowd behavior assessment criteria. A numerical range has been defined for each type of evaluation criterion. Within this range, the target evaluation values can be entered into the crowd MDoF behavior static state optimization module either by dragging a slider or by entering the values directly.

3.5.2. Group clustering

Predefined mode clustering. In the predefined clustering approach, a specific distance function is chosen, for which the user specifies key parameters and the desired number of clusters. Clustering is performed using the K-means algorithm [42], where each individual i is assigned a label $G(i)$. The distance function value of each individual can be assigned to the spatial ordering value O_i after normalization. Considering the spatial relationships between the crowd members, various clustering modes can be pre-established, including unidirectional distance, surrounding angle, concentric distance, and contour distance clustering. Examples of the four clustering methods are illustrated in Fig. 5.

User sketch clustering. To increase user autonomy in clustering methods and inspired by CrowdBrush [14], we developed a brush tool that allows users to directly sketch the spatial locations of a crowd to assign group labels, $G(i)$. The brush, configurable with a specific label, G_{brush} , automatically assigns $G(i) = G_{brush}$ to individuals within its reach. Furthermore, the order in which individuals are grouped using this tool is documented and normalized, contributing to each individual's spatial ordering value, O_i .

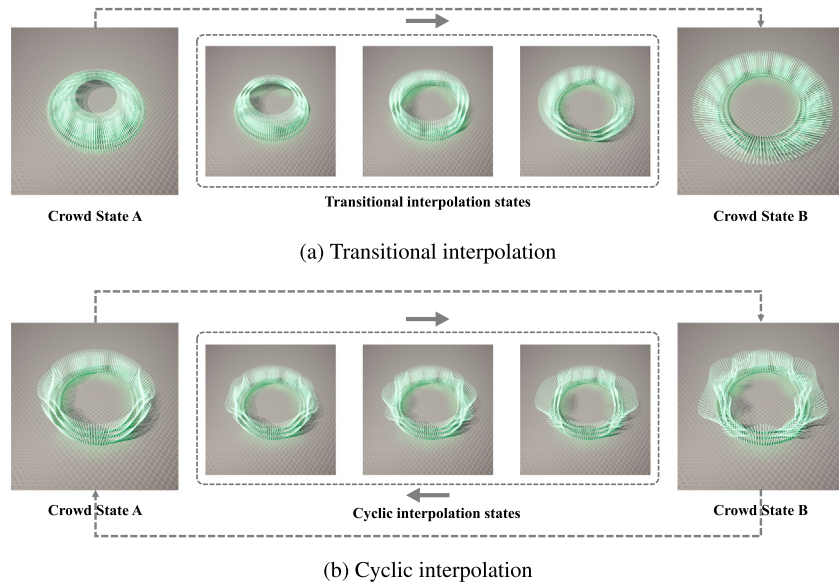


Fig. 4. Examples of the interpolation process between MDoF states of a crowd.

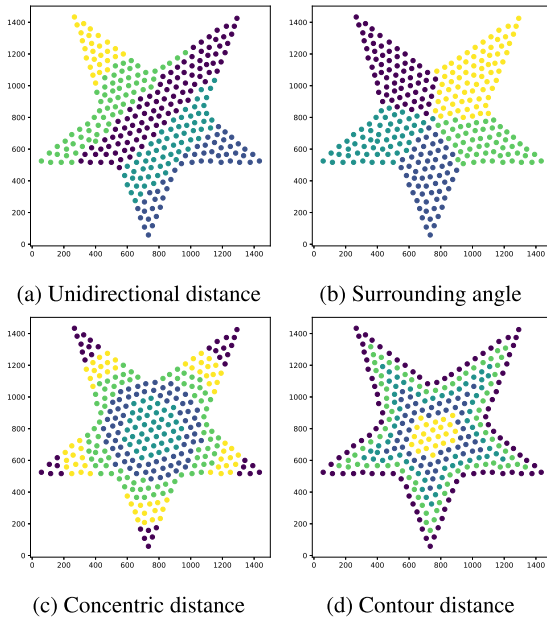


Fig. 5. Group clustering of a pentagram-shaped crowd, utilizing four different distance functions and the K-means algorithm with $k = 5$. The representation employs different colors to indicate different clustering groups.

The brush tool offers enhanced functionality through three interaction modes: translation, scaling, and rotation, extending the scope of user-generated sketches. In addition, the proposed interaction methods also record the drawing order to further enhance the agents' behavior by incorporating not only spatial position shapes but also time information. Fig. 6 demonstrates the labeling process using these modes.

3.5.3. Custom spline tool

Custom spline functions allow users to interactively define complex functions, useful as time offset functions (TO) or interpolation control functions (Lerp) in state interpolation modules. The custom spline function tool allows easy modification of function shapes, enhancing the variability of MDoF behaviors in crowd control. The tool supports

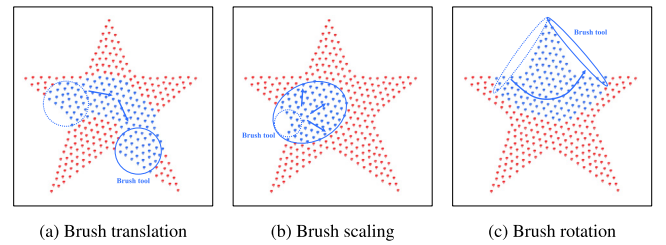


Fig. 6. Group clustering in a pentagram-shaped crowd was facilitated through user sketches, employing three distinct brush interaction modes.

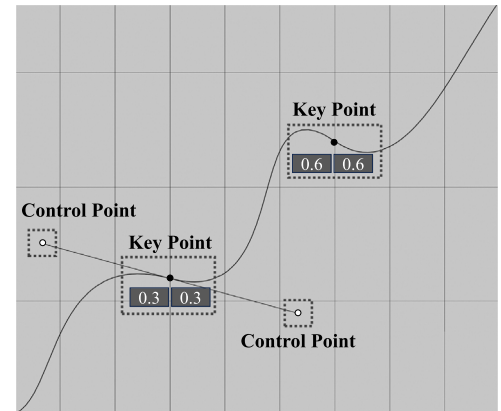


Fig. 7. The interactive interface of the custom spline function.

modifications through numerical input or direct manipulation of key points, and allows users to adjust the magnitude and direction of tangents at control points. This interaction is illustrated in Fig. 7.

The interaction methods of User Sketch Clustering and the Custom Spline Tool are demonstrated in the video included in the supplementary materials.

4. Results and discussion

In this section, we demonstrate the application of our framework to managing MDoF behaviors in a 393-person crowd using props in

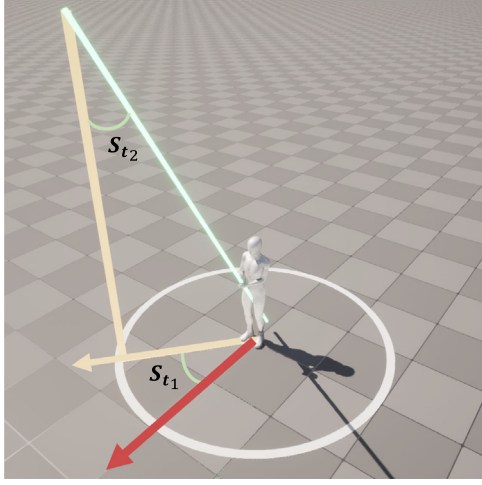


Fig. 8. The target behavior space of an individual.

a multi-dimensional behavior space. We begin by defining standard behavioral patterns and evaluating the effectiveness of state generation optimization algorithm, focusing on the optimization metrics discussed in Section 3.3.2 and their impact on crowd state planning. We also evaluate the results of the generated crowd states under various constraints and explore the influence of spatio-temporal ordering techniques, along with TO and Lerp functions, during the interpolation phase. Finally, we compare the simulated outcomes with the actual performance observed during the rehearsal at the Beijing Winter Olympics, highlighting the practical effectiveness of our framework. The simulations are visualized using Unreal Engine on a PC with an Intel(R) Core(TM) i9-10900KF CPU @ 3.70 GHz, 64.0 GB RAM, and an NVIDIA GeForce RTX 3090.

For the 393-person crowd performance, we used their positions as the constrained behavioral space, driven by the formation data. The orientation and amplitude of their props are set as the target behavioral spaces, denoted as S_{t1} and S_{t2} . The value ranges for these spaces are $[0, 360]$ and $[-90, 90]$ respectively, with both measured in degrees. Fig. 8 shows the target behavior space for each individual in the crowd.

4.1. Crowd MDoF static state optimization

Fig. 9 shows the visualization results of two crowd state optimization examples using *Unidirectional Distance* and *Concentric Distance* as clustering methods, with clustering labels integrated as constraints in the optimization process. As the optimization algorithm progresses through 200 iterations, the behavioral differences within groups diminish, resulting in similar optimized shapes and spatial distributions

Table 2
Configures of target evaluations in 2 example optimizations.

Evaluation	Optim 1	Optim 2
$MI(S_{loc} \text{ and } S_{t1})$	1.51	3.35E-1
$MI(S_{loc} \text{ and } S_{t2})$	\	2.52E-1
$MI(S_{t1} \text{ and } S_{t2})$	2.23E-3	\
Spatial continuity	9.49E+2	2.54E+3
Entropy (S_{t1})	\	3.47
Entropy (S_{t2})	1.81	\

of group labels. The configurations of the target parameters for both examples are detailed in Table 2, and the convergence of the evaluation criteria is depicted in Fig. 10.

In Section 3.3.2, we introduced three evaluation metrics (mutual information, spatial continuity, and entropy) to refine targeted MDoF static behavior states within a crowd. To evaluate the effectiveness of these metrics, we conducted three different experiments, each designed to assess their impact on optimizing the target states of the crowd.

Fig. 11(a) shows the impact of varying target mutual information (MI) values on the optimized target states of a crowd, by analyzing the orientation distribution (S_{t1}) of props held by 393 agents and their spatial position. At a target $MI = 0$, prop orientations are predominantly to the right, indicating minimal spatial correlation with the spatial positions, which are uniformly distributed. At a target of $MI = 0.75$, orientations are aligned in two primary directions, indicating slight variation. At a target of $MI = 1.78$, prop orientations are uniformly distributed, suggesting a strong correlation between the two behavior dimensions. This demonstrates how different MI values influence the interrelationships between the different dimensions of crowd behavior.

Fig. 11(b) demonstrates how different values of the target spatial continuity influence the optimized state of a crowd, treated as a single cluster. At $C_{smooth_goal} = 0$, the characteristics of the behavioral space, such as the angle and direction of the props, are uniform with low variability. As C_{smooth_goal} increases, there is a noticeable increase in the variance of the orientation and amplitude of adjacent props. This shows that adjusting spatial continuity values can effectively control inter-individual relationships within localized areas of the crowd.

Fig. 11(c) shows the effect of different values of target entropy on the crowd's optimized target behavior, focusing in particular on the amplitude distribution of props with a constant orientation. At a target entropy of $C_{entropy_goal} = 0$, the amplitude of the props remains fairly uniform, contributing to an overall organized visual appearance. As $C_{entropy_goal}$ increases, the amplitude distribution among the props becomes increasingly varied, resulting in a visually more chaotic scene. This demonstrates that higher entropy values significantly enhance scene complexity.

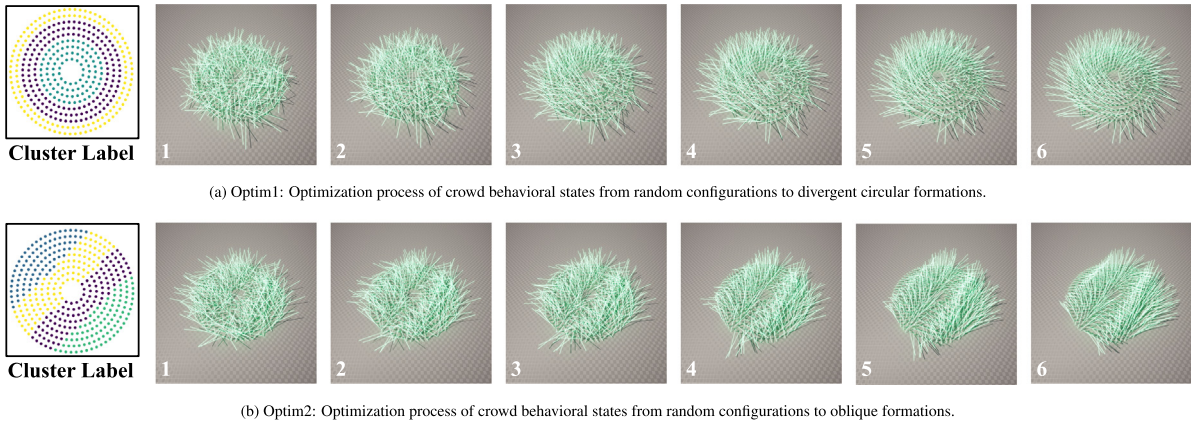


Fig. 9. Examples of the optimization process of static MDoF states of a 393-person crowd.

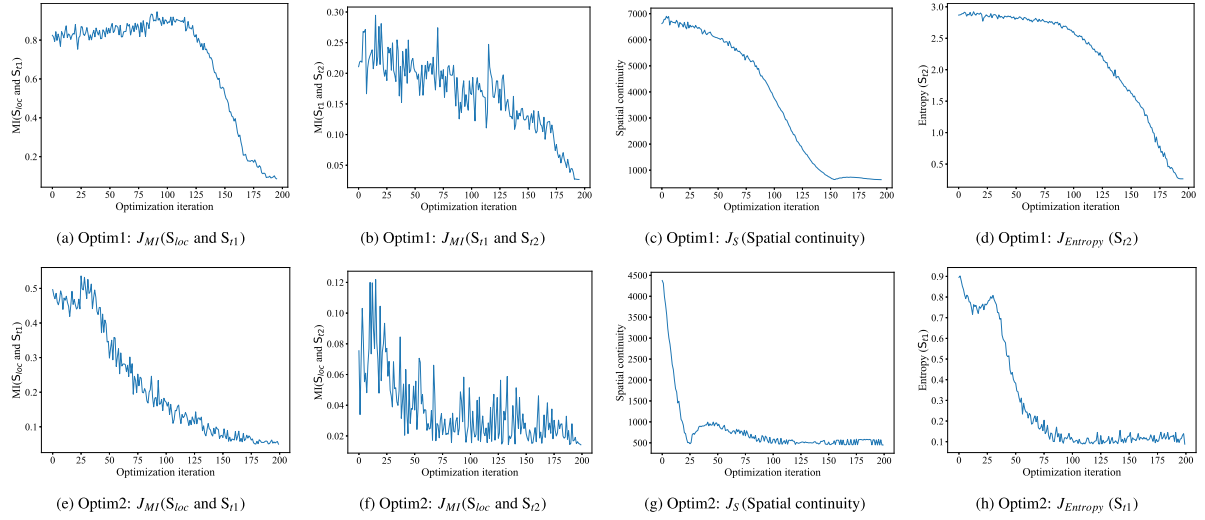


Fig. 10. Optimization of the objective functions during the two example processes for crowd state optimization.

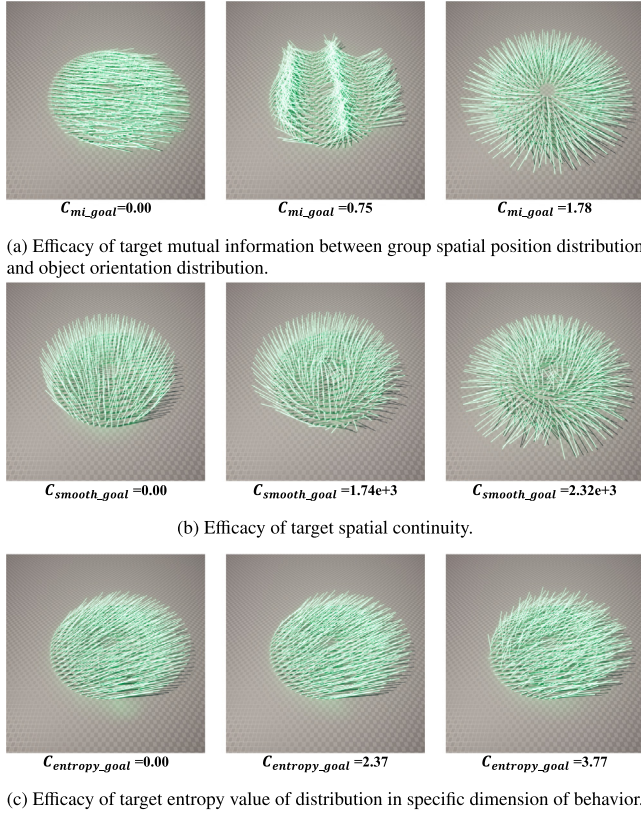


Fig. 11. Examples of the optimization process of static MDoF states of a 393-person crowd.

4.2. Crowd MDoF dynamic interpolation

Our module for interpolating MDoF crowd states facilitates transitions between states and enhances diversification of crowd behavior through spatio-temporal ordering, using the time offset function TO and the interpolation progress control function Lerp. We used a 393-person crowd in three concentric circles with two predefined crowd states, S_A and S_B , for interpolation, as shown in Fig. 12. We pre-assign the group clustering labels $G(i)$ based on the concentric distance function and spatial ordering value O_i using the normalized surrounding angle ordering values. Fig. 13 shows the visualization of the group clustering and spatial ordering of the crowd.

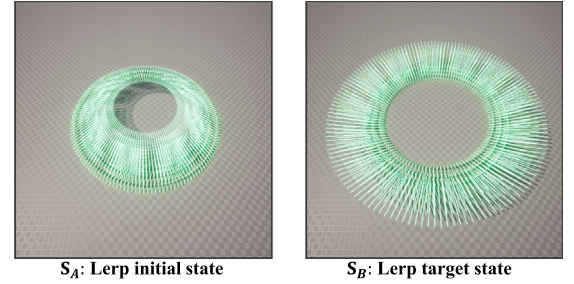


Fig. 12. The example initial and target states of the dynamic interpolation.

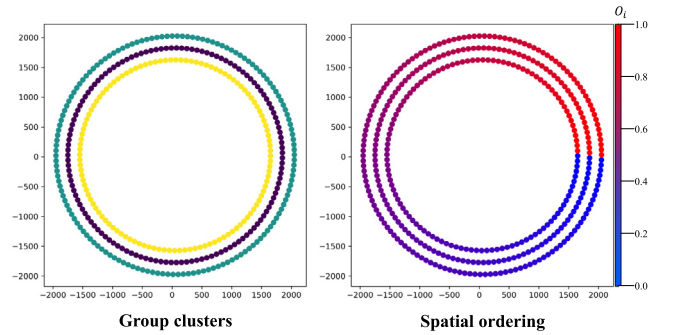


Fig. 13. Pre-settings of group states interpolation. The left side shows grouping into three categories based on distance from the center, represented by different colors for group labels. On the right side, individuals are ordered based on their circumferential angles, with sorting values ranging from 0 to 1.

To compare the impact of different time offset functions TO on crowd behavior, we set Lerp as a linear function and the interpolation time for each individual to $T = 5$ s. Additionally, we established a set of different TO functions to compare the behavioral states of the crowd during the interpolation process.

Fig. 14 shows the impact of five different time offset functions on the transition state from S_A to S_B . TO1 is a constant function, meaning all individuals start interpolating their states simultaneously, and all complete the transitional interpolation by $t = 6$ s. In TO2, the time offset value increases rapidly with the ordering value and then slows down. In terms of crowd behavior, this means that only a few individuals with lower O_i values have lower time offsets, and only these few start the transition between 2–5 s. From the $t = 6$ s on, most individuals start interpolating states rapidly one after the other,

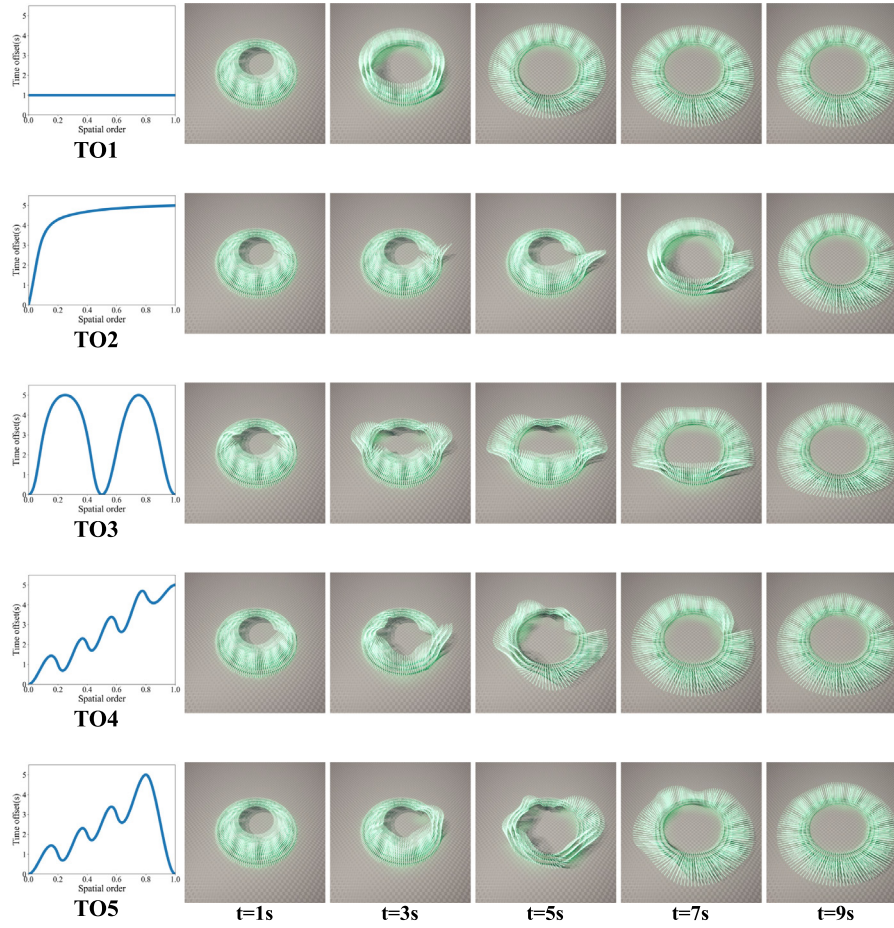


Fig. 14. Comparison of the effects of 5 Time Offset functions on the interpolation process of crowd states.

achieving a controlled rhythm in the overall crowd state interpolation. TO3, TO4, and TO5 demonstrate the influence of different custom functions on the intermediate process of crowd state interpolation. The shape of the function is directly correlated with the appearance of the crowd state, allowing the user to intuitively adjust the intermediate states. Both TO3 and TO5, which have equal values at both ends of their domains, can produce continuous intermediate transition states in circular formations. Different time offset functions demonstrate the extensibility of the interpolation module based on spatio-temporal ordering, which is able to generate diverse crowd behaviors.

In addition to the time offset function, the interpolation progress control function Lerp can also adjust the behavior of the crowd during the interpolation process. We fixed the time offset function TO3 and created several Lerp functions to compare the behavior of the crowd in different interpolation states.

Fig. 15 shows the impact of four different interpolation progress control functions and their combinations on the transition process from the crowd state S_A to S_B . Lerp1 is a smoothed linearly increasing function, starting with a value of 0 at the beginning of the domain and ending with a value of 1, indicating a smooth transition of the crowd from state S_A to S_B . This represents the transitional interpolation mentioned in Section 3.4.2. Lerp2, Lerp3, and Lerp4 have the same function values at the start and end points of their domains, corresponding to cyclic interpolation. Comparing Lerp2 and Lerp3, it can be observed that the periodicity of the Lerp function forms repetitive patterns in the appearance of crowd behavior, and the amplitude of the function also affects the extent of the shapes exhibited in crowd behavior. In addition, by applying different Lerp functions to various clustered groups within the crowd, more diverse crowd behavior can be achieved, as illustrated in the fourth row of Fig. 15.

4.3. Computation time

Computational time in our framework is divided into two segments: optimization time for static crowd behavior states, and computation time for dynamic states during interpolation. The optimization time is influenced by the number of agents, as demonstrated in extensive experiments showing varied crowd sizes from 20 to 390 agents across 200 optimization iterations (see results in Fig. 16, where each value is the average of 20 experiments with the same configuration). It can be seen that regardless of the number of individuals, the optimization time cannot achieve real-time computation. However, since static behavior states are pre-computed, real-time computation is not required for this part. We only need to maintain real-time computation and interaction during the interpolation process.

The computational cost for dynamic interpolation of MDoF behaviors is related to their complexity. For example, the system achieved a frame rate of 90 fps, for the 393-agents prop performance, and 60 fps for the more complex 306-agents skeletal animations, both ensuring real-time interaction.

4.4. Applicability experiments

To validate the applicability of our framework, we extended it to the predefined choreography of group actions. We set the target degree of freedom space to the interpolation alpha value of the animation blend space and used character animation resources from the Mixamo online platform [43] as static behaviors in the MDoF behavior space. The group behaviors demonstrated under the influence of the TO function and the Lerp function are shown in Fig. 18.

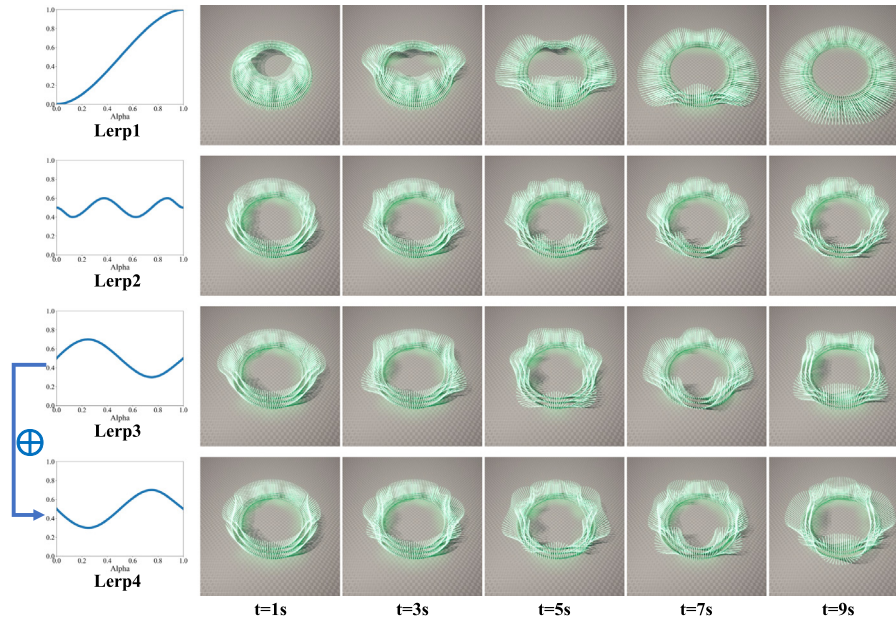


Fig. 15. Comparison of the effects of 4 Lerp functions on the interpolation process of crowd states.

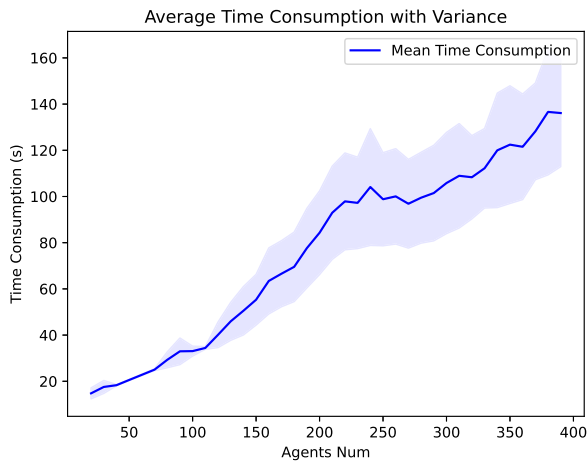


Fig. 16. Optimization computation time for static behavior states across different agent numbers. The blue line represents the average computation time, while the light blue area shows the range within a standard deviation from the mean.

We have applied our proposed method and system in actual crowd performances, including the *Beginning of Spring* segment in the Opening Ceremony of the 2022 Beijing Winter Olympics. In the *Beginning of Spring* segment, a total of 5 regular patterns were presented. Fig. 17 shows the comparison of spatial group clustering, crowd behavior planning simulation and actor rehearsals of the actual performance for these 5 performance patterns.

The left side of Fig. 17 shows the pre-clustered groups for each scene, the middle section displays the simulations for each performance segment, and the right side shows the actual actor rehearsals. The choreographers used our method and system to pre-plan the performance scheme before rehearsals and then guided the real actors to rehearse and perform according to these standards. This use case demonstrates the practicality and real-world applicability of our proposed method. Dynamic scenes from these two application examples can be found in the supplementary video materials.

5. Conclusion

This research models the MDoF behavior of a crowd and presents a framework for planning such behaviors using optimization and spatio-temporal ordering. By employing metrics such as mutual information, spatial continuity, and entropy across different dimensions of the crowd's behavior, we successfully optimize the static MDoF states of the crowd. The use of spatio-temporal ordering and various user interaction designs has facilitated diverse behavior during state transitions within the crowd. The effectiveness of this approach in optimizing crowd states and enhancing behavioral variety is corroborated by experimental results. Both user studies and real-world applications have confirmed the practicality of our methodology.

Future research will explore optimization strategies for MDoF crowd behaviors, adhering to the No Free Lunch (NFL) theorem. We plan to investigate and benchmark the effectiveness of different algorithms in crowd behavior planning and to develop complex behaviors such as synchronized dancing at a crowd scale.

CRedit authorship contribution statement

Kexiang Huang: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Gangyi Ding:** Supervision, Project administration. **Dapeng Yan:** Data curation. **Ruida Tang:** Visualization. **Tianyu Huang:** Supervision. **Nuria Pelechano:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for *[Computers & Graphics]* and was not involved in the editorial review or the decision to publish this article.

Data availability

Data will be made available on request.

Acknowledgments

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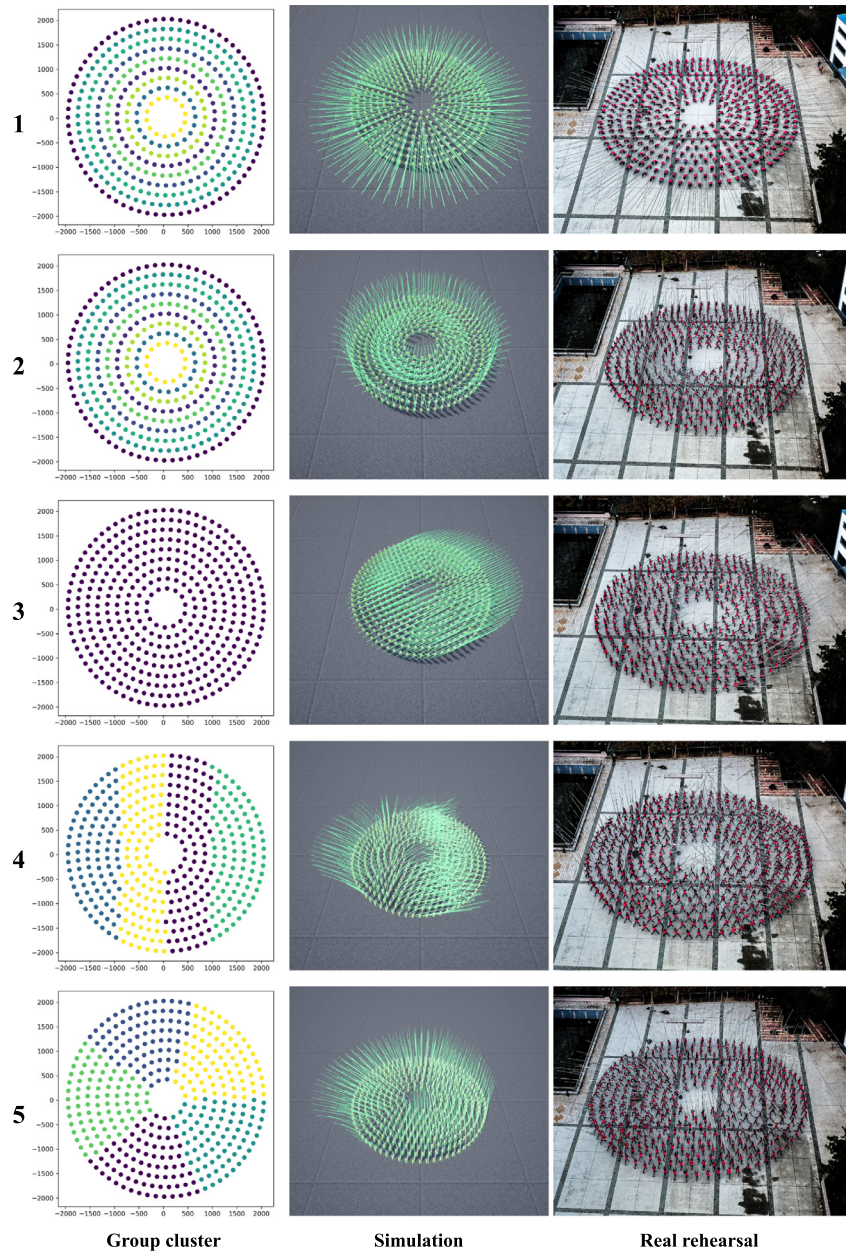


Fig. 17. The application of crowd multi degree of freedom behavior planning method in the rehearsal of the Winter Olympics Opening Ceremony performance for the *Beginning of Spring*.

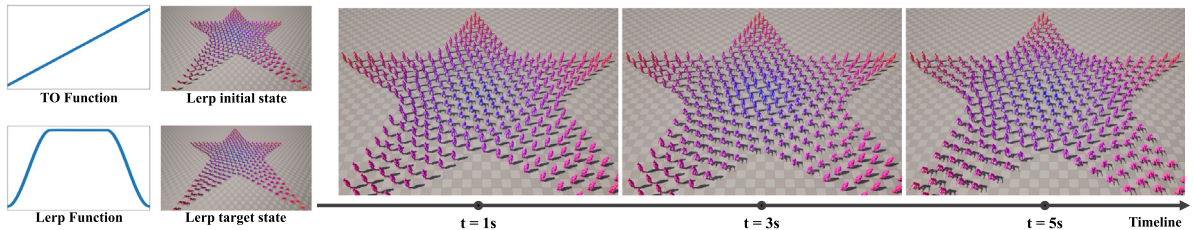


Fig. 18. Dynamic interpolation of MDoF behavior in a 306-person crowd arranged in a star shape, transitioning from an initial state of idle standing to a target state of squatting. Agent colors represent the spatial-ordering value, with blue to red indicating a range from 0 to 1.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cag.2024.104051>.

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